

Roshan Raj Mahato

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SUMMARY

Machine Learning Engineer with hands-on experience building **end-to-end DL models**, **RAG pipelines**, and **LLM-powered applications** using **PyTorch**, **TensorFlow**, and **LangChain**. Proficient in **deploying ML solutions on cloud infrastructure (AWS, Docker)** and designing **GenAI workflows** with **OpenAI GPT-4** and **HuggingFace**. Seeking to leverage **ML/DL expertise** at **Lytix** to build scalable, production-ready AI systems that protect people on the road.

TECHNICAL SKILLS

Languages: Python (NumPy, Pandas, Scikit-learn, Matplotlib), C, C++, SQL
Deep Learning: PyTorch, TensorFlow, Keras, CNNs, RNNs, ANNs, **Transfer Learning** (VGG16)
Machine Learning: Supervised/Unsupervised Learning, XGBoost, Random Forest, SVM, PCA, K-Means, **Hyperparameter Tuning**, **Feature Engineering**, **Cross-Validation**, **Regularization**
GenAI & LLMs: LangChain, RAG Pipelines, Prompt Engineering, OpenAI GPT-4, HuggingFace Transformers, FAISS, Llama-2, **AI Agents**
MLOps & Deployment: Docker, ONNX, MLflow, AWS (EC2, ECR), GCP (Cloud Functions), Git, GitHub
Data & Databases: MySQL, Oracle SQL, Power BI (DAX), Microsoft Excel

EDUCATION

Ramgarh Engineering College <i>Bachelor of Engineering – Computer Science and Engineering CGPA: 7.0/10</i>	Ramgarh, Jharkhand <i>Jun 2021 – Jul 2025</i>
D.A.V Public School, Moonidih <i>Intermediate – Class XII 89%</i>	Dhanbad, Jharkhand <i>Jun 2018 – Jul 2020</i>

EXPERIENCE

Machine Learning Intern – Credit Card Fraud Detection Python, Logistic Regression, Classification <i>YBI Foundation</i>	Jul 2024 <i>Remote</i>
- Developed a Credit Card Fraud Detection model using logistic regression on imbalanced transaction data (284K+ transactions), achieving 87% accuracy in distinguishing fraudulent from genuine activities. - Analyzed transaction patterns to identify anomalies through feature engineering and statistical methods, improving model precision on rare fraud cases by 18%. - Optimized detection efficiency via threshold tuning and evaluation metrics (precision, recall, F1-score), contributing to prevention of financial losses in real-world deployment scenarios.	

PROJECTS

Student Performance Prediction Python, Scikit-learn, Regression, Classification, MLflow	Dec 2023
- Engineered and deployed an end-to-end supervised ML pipeline achieving 96% prediction accuracy on a dataset of 10K+ student records for academic performance forecasting. - Improved data quality by 33% through systematic missing value imputation, categorical encoding, and feature engineering, reducing model error on unseen data. - Benchmarked 6+ classification and regression algorithms (XGBoost, Random Forest, Logistic Regression) using cross-validation to select optimal model; tracked experiments with MLflow.	
Pneumonia Detection & Classification Python, PyTorch, TensorFlow, Keras, VGG16, CNN	Sept 2022
- Trained a CNN leveraging VGG16 Transfer Learning to classify chest X-ray images (5,863 images, 2 classes) as Pneumonia or Normal, achieving 91% test accuracy. - Applied image augmentation and preprocessing techniques (rotation, flipping, normalization) to improve model robustness, reducing overfitting and improving generalization on real-world medical imaging data.	
Legal AI Assistant Python, LangChain, FAISS, OpenAI GPT-4, HuggingFace, Docker	Jan 2026
- Built a RAG-based Legal AI Assistant that parsed PDF/DOCX contracts at scale, extracting clauses and performing automated risk scoring (HIGH/MEDIUM/LOW) with 92% accuracy using GPT-4 and LangChain. - Implemented FAISS vector store with HuggingFace embeddings for semantic search, enabling grounded Q&A with mandatory section citations and reducing manual contract review time by 60%. - Designed modular, production-ready pipeline with dedicated components for summarization, risk analysis, and contradiction detection – deployable via Docker for consistent cross-environment execution.	